

Nivedhitha Duggi

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Education

University of North Texas, MS in Computer and Information Science Expected May 2026
Achievements: Distinguished Graduate Student for the 2024–2025 academic year

Rajiv Gandhi University of Knowledge Technologies, B.Tech in CS Nov 2020 – Apr 2024
Achievements: Dean's Merit Scholarship in Fall 2022, and Spring 2023

Experience

Software Engineer Intern, Lennox International Inc – Carrollton, Tx May 2025 – Present

- Developed a GenAI-based customer notification platform for residential HVAC dealers, integrating Calendly APIs, Azure Functions, and OpenAI's LLM with RAG. Improved customer engagement by reducing 62% unanswered calls and lowering voicemail reliance from 3%, directly mitigating revenue loss.
- Engineered and Deployed automated text and table extraction pipelines for HVAC schedules and developed AI-driven Power Apps with RAG-based test automation using Python, Azure Cognitive Services, Azure Functions, and OpenAI APIs reducing quote preparation time from 3 days to few hours and cutting manual scheduling time& report generation by 60% across lab operations.
- Conducted S40 Thermostat ELC analysis, IFC board FAI, and flame-sensing tests on 17K+ returned units using Python and Excel automation, improving hardware validation accuracy by 35% and reducing repeat field failures by 22% across HVAC systems.

Graduate Research Assistant, University of North Texas – Denton, TX Oct 2024 – May 2025

- Evaluated GLUE and SuperGLUE benchmarks for ClinicalBERT (clinical notes/discharge summaries) and PubMedBERT (biomedical literature); analyzed fairness/bias and proposed domain-specific benchmark adoption to improve reliability for clinical decision-support (COHB Lab, Prof. Mohammed Aledhari).
- Developed LLM pipelines to generate MCQs and clinical assessments from biomedical data using GPT-style models fine-tuned on MIMIC-III; validated with MLMU and ARC to improve quality and clinical relevance of AI-driven assessment tools (Biomedical AI Lab, Prof. Sagnik Ray Choudhury).
- Benchmarked DNN/ML inference and video transcoding on heterogeneous cloud platforms; analyzed latency/throughput/energy and co-authored a paper on workload optimization strategies (HPCC Lab, Prof. Mohsen Amini Salehi).

Python Developer Intern, Exponential AI – Hyderabad, IN April 2024 – Aug 2024

- Contributed on the Solutions Team to develop scalable ML pipelines for healthcare insurance products, leveraging Random Forest and Deep Learning for contract data extraction, boosting throughput and reliability by 49%.
- Applied BERT, fine-tuned DistilBERT, and RoBERTa for extractive QA on scanned documents to extract sensitive keywords, integrated Llama and Mistral for nuanced information retrieval with NLTK and spaCy and tuned YOLOv8 for precise handwritten recognition. Implemented RAG and agentic workflows to improve data-driven decisioning and increase keyword-extraction accuracy.

Systems Engineer Intern, Pragyashal – Chennai, IN Jan 2024 – Apr 2024

- Developed Progressive Web Apps (PWAs) for Android and iOS using React.js, Next.js, and MongoDB, optimizing response times and ensuring cross-platform accessibility for an immersive VR education platform.
- Implemented Micro Frontend (React + Angular) and SEO strategies, enhancing scalability and online visibility.

Data Analyst Intern, Pine Labs – Bengaluru, IN May 2023 – Aug 2023

- Developed a Python-based NPV profitability tool (NPV, risk-adjusted yield, ROI) across product tiers, identifying high-value segments and contributing to \$1M+ in additional annual revenue.
- Built automated PowerBI dashboards (Finance, Macroeconomics, MBR) and designed A/B test pilots (income verification, retail), saving 20+ reporting hours/month and delivering a 10% conversion lift of booked loans.

Software Engineer Intern, Achala IT Solutions Pvt Ltd – Hyderabad, IN

May 2022 – Aug 2022

- Revitalized and restored SwiftLint in the CI pipeline, integrating Git Hooks to enforce coding standards and reduced linting time by 83.3%.
- Improved development efficiency and code reliability by automating style checks, reducing manual review effort, and ensuring consistent, maintainable code across teams.

Technical Skills

Programming: Python, C/C++, Java, JavaScript, SQL, Data Structures and Algorithms, OOP

Web & Frameworks: React, Next.js, Node.js, Express.js, Django, Flask, FastAPI, HTTP, TCP/IP, DNS, TLS.

Machine Learning & AI: Regression, Classification, Clustering, Scikit-Learn, TensorFlow, PyTorch, XGBoost, Hugging Face Transformers, LangChain, CNN, RNN

NLP & LLMs: OpenAI API, Transformer Models (BERT, DistilBERT, RoBERTa), RAG Systems

MLOps & Deployment: MLflow, Weights & Biases, DVC, Kubeflow, ONNX, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), FastAPI serving

Data & Analytics: Pandas, NumPy, Matplotlib, PowerBI, Tableau, Apache Spark, Hadoop, Airflow, ETL Pipelines, Statistical Analysis (Hypothesis Testing, A/B Testing, Model Evaluation)

Cloud Platforms: AWS (EC2, S3, Lambda, SageMaker, Redshift), Microsoft Azure, Google Cloud (GCP)

Databases & Warehousing: MySQL, PostgreSQL, MongoDB (NoSQL), Snowflake

Version Control & Systems: Git, GitHub, Linux

Projects

Natural Language Processing with Disaster Tweets

- Developed a logistic regression model on 10K tweets with preprocessing (tokenization, stemming, TF-IDF), achieving 80.3% accuracy.
- Addressed class imbalance with RandomOverSampler and visualized results using Seaborn and Matplotlib.
- **Technologies:** Python, Logistic Regression, Scikit-learn, NLTK, Pandas, Matplotlib

Connect4 Reinforcement Learning

- Built an agent for Connect4 game using Q-Learning and Deep Q-Learning, optimized with OpenAI Gym and Stable-Baselines3.
- Improved gameplay robustness by optimizing reward functions and exploration strategies, leading to consistently superior performance against AI opponents.
- **Technologies:** Reinforcement Learning, Q-Learning, Deep Q-Learning, Python, OpenAI Gym

Twitter Clone – Connect

- Developed a full-stack Twitter clone with secure authentication, dynamic post management, and a responsive UI replicating core social media features.
- Implemented RESTful APIs with Node.js/Express.js and integrated MongoDB for real-time data storage and retrieval, ensuring scalable backend performance.
- **Technologies:** HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Git

Recycrafter: AI-Powered Gamified Recycling Platform

- Developed an interactive 2D game using Python and Pygame, featuring multi-level progression and real-time obstacle avoidance mechanics.
- Reduced rendering latency by 25% while integrating visuals, music, and smooth transitions.
- **Technologies:** Python, Pygame, Git/GitHub, Object-Oriented Programming

Leadership & Awards & Certifications

RCOB/MGMT Ken & Shirley Murphy Scholarship, University of North Texas

May 2025

MLH Best Use of Auth0, HackUNT 2024

Dec 2024

Member, Women in Computing, University of North Texas

Nov 2024 – Present